

# CS 237: PROBABILITY IN COMPUTING

INSTRUCTORS: TIAGO JANUARIO, NATHAN KLEIN

**BOSTON**  
UNIVERSITY

© MARK ANDERSON, WWW.ANDERSTOONS.COM



"I wish we hadn't learned probability 'cause I don't think our odds are good."

## LECTURE 2

### Last time

- Course information
- Introduction to probability theory
- Sample Spaces and Events
- Examples

### Today

- Probability function
- Symmetry
- Probability axioms

## HOMEWORK SUBMISSION

- ▶ Deadline: Thursdays, at 09:00 PM
- ▶ Register at <https://www.gradescope.com/courses/1226187>
- ▶ Use entry code **N22E5K**
- ▶ Sign up using your BU email, and include your Student ID in the format U000000000
- ▶ Use Gradescope for regrades as well



## RECAP: PROBABILITY CAST OF CHARACTERS

- ▶ **Experiment:** a repeatable procedure
- ▶ **Outcome:** result of the experiment
- ▶ **Sample space  $\Omega$ :** set of all possible outcomes
- ▶ **Event:** a subset of the sample space

# PROBABILITY: CAST OF CHARACTERS

- ▶ **Experiment:** a repeatable procedure
  - ▶ Toss a coin  $\Omega = \{H, T\}$
  - ▶ Toss a coin 3 times  $\Omega = \{HHH, HTH, \dots\}$
  - ▶ Roll two dice  $\Omega = \{(1, 2), (2, 5), \dots\}$
  - ▶ Shuffle a deck of cards  $|\Omega| = 52!$  possible permutations
  - ▶ Draw a sock from a drawer containing red and black socks; if the sock is red, draw another sock

## PROBABILITY: CAST OF CHARACTERS

- ▶ **Outcome:** result of the experiment
- ▶ **Sample space  $\Omega$ :** set of all possible outcomes
  - ▶ **Toss a coin** *let  $H = \text{"heads"}$  and  $T = \text{tails}$*

$$\Omega = \{H, T\}$$

## PROBABILITY: CAST OF CHARACTERS

- ▶ **Outcome:** result of the experiment
- ▶ **Sample space  $\Omega$ :** set of all possible outcomes
  - ▶ Toss a coin 3 times

$$\Omega = \{ HHH, HTH, THH, \dots, TTT \}$$

$$|\Omega| = 8$$

## PROBABILITY: CAST OF CHARACTERS

- ▶ **Outcome:** result of the experiment
- ▶ **Sample space  $\Omega$ :** set of all possible outcomes
  - ▶ Roll two 4-sided dice

$$\Omega = \{(i, j) \mid i, j \in \{1, 2, 3, 4\}\}$$

$$|\Omega| = 4 \cdot 4 = 16$$

$$\Omega = \{(3, 4), (2, 1), (1, 2), \dots\}$$

Hand-drawn table representing the sample space for rolling two 4-sided dice.

|         |   | 2nd die |       |   |   |
|---------|---|---------|-------|---|---|
|         |   | 1       | 2     | 3 | 4 |
| 1st die | 1 | (1,1)   | (1,2) |   |   |
|         | 2 | (2,1)   |       |   |   |
|         | 3 |         |       |   |   |
|         | 4 |         |       |   |   |



## PROBABILITY: CAST OF CHARACTERS

- ▶ **Outcome:** result of the experiment
- ▶ **Sample space  $\Omega$ :** set of all possible outcomes
  - ▶ Roll a 4-sided die twice

1st roll  
2nd roll

$$\Omega = \{(i, j) \mid i, j \in \{1, 2, 3, 4\}\}$$

$$|\Omega| = 4 \cdot 4 = 16$$

$$\Omega = \{(3, 4), (2, 1), (1, 2), \dots\}$$

2nd roll

|          | 1 | 2 | 3 | 4 |
|----------|---|---|---|---|
| 1st roll |   |   |   |   |
| 1        |   |   |   |   |
| 2        |   |   |   |   |
| 3        |   |   |   |   |
| 4        |   |   |   |   |



## PROBABILITY: CAST OF CHARACTERS

- ▶ **Outcome:** result of the experiment
- ▶ **Sample space  $\Omega$ :** set of all possible outcomes
  - ▶ Toss a coin until we get a heads

$$\Omega = \{H, TH, TTH, TTTT, \dots\}$$

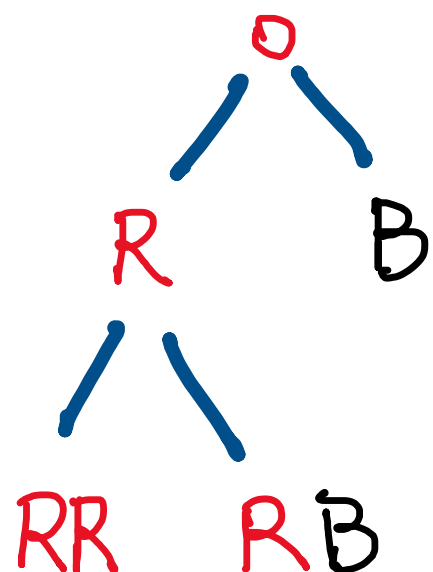
## PROBABILITY: CAST OF CHARACTERS

- ▶ **Outcome:** result of the experiment
- ▶ **Sample space  $\Omega$ :** set of all possible outcomes
  - ▶ Shuffle a deck of cards and display all cards

$$|\Omega| = 52!$$

## PROBABILITY: CAST OF CHARACTERS

- ▶ **Outcome:** result of the experiment
- ▶ **Sample space  $\Omega$ :** set of all possible outcomes
  - ▶ Draw a sock from a drawer containing red and black socks; if the sock is red, draw another sock



$$\Omega = \{B, RR, RB\}$$

## PROBABILITY: CAST OF CHARACTERS

- ▶ **Event:** a maximal subset of the sample space
  - ▶ Experiment: toss a coin 3 times
  - ▶ Event: get at least 2 heads (2 or more heads)

$$E = \{HHH, HHT, HTH, THH\}$$

↑  
3 heads

2 heads

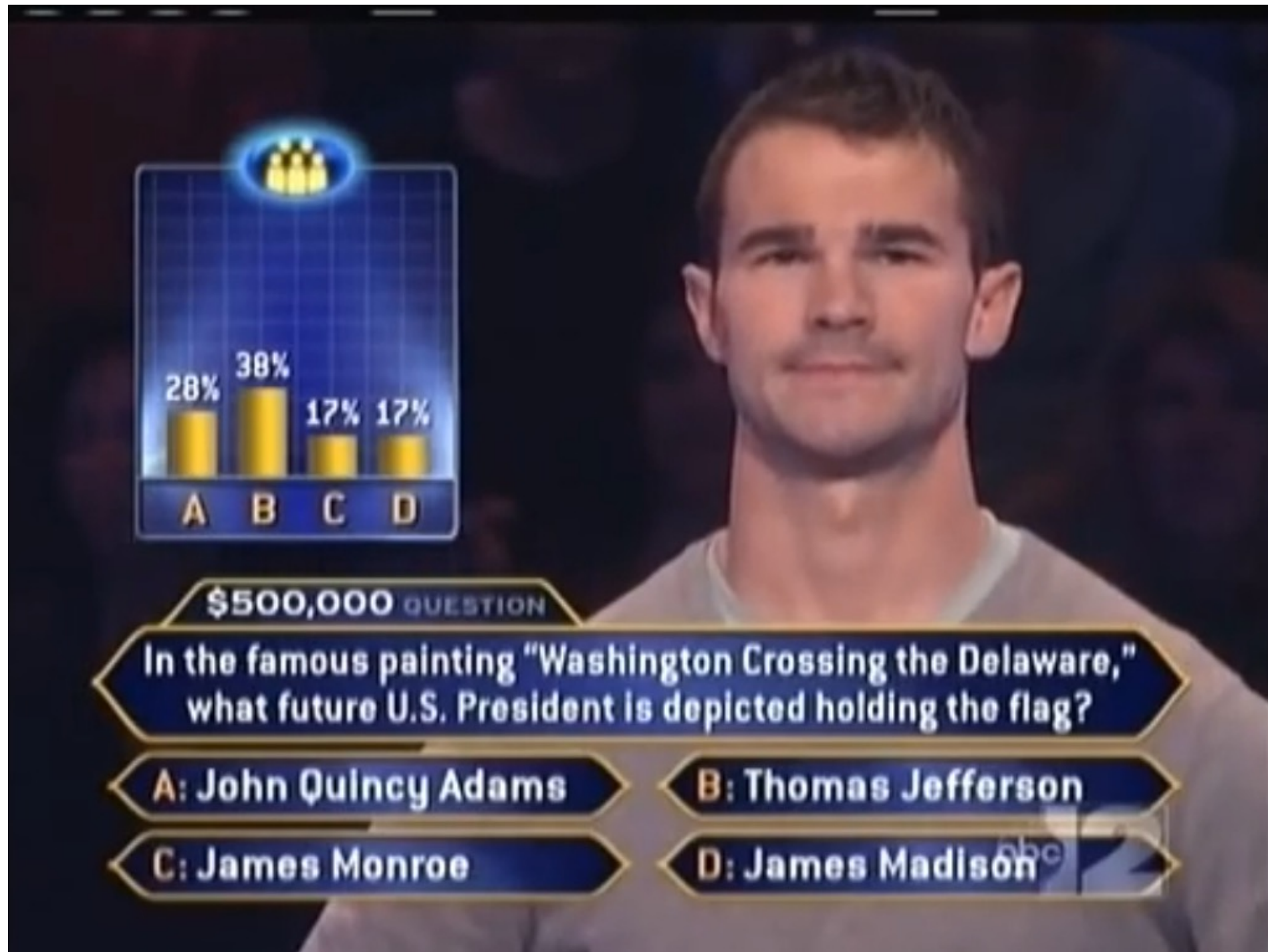
## PROBABILITY: CAST OF CHARACTERS

- ▶ **Event:** a maximal subset of the sample space
- ▶ Experiment: roll two 6-sided dice
- ▶ Event: the sum of the two numbers rolled is 11

$$\Omega = \{(5, 6), (6, 5)\}$$

1st die      2nd die

## ASK THE AUDIENCE



## ASK THE AUDIENCE

- Experiment: toss a coin 3 times

$$A \cap B = \emptyset$$

Are the following events disjoint?

$A$  = "exactly 2 heads"

$B$  = "exactly 2 tails"

A. YES

B. NO

It is not possible to have  
two heads and  
two tails



## LET'S VOTE



- Experiment: toss a coin 3 times

$$E_1 \subseteq E_2$$

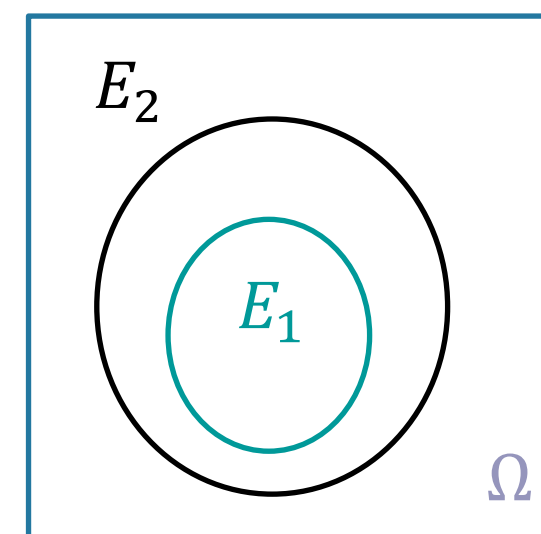
Does the first event **imply** the second?

$E_1$  = "at least 2 heads" = {HHT, HTH, THH, HHH}

$E_2$  = "exactly 2 heads" = {HHT, HTH, THH}

A. YES

B. NO





## RECAP: PROBABILITY CAST OF CHARACTERS

- ▶ **Experiment:** a repeatable procedure
- ▶ **Outcome:** result of the experiment
- ▶ **Sample space  $\Omega$ :** set of all possible outcomes
- ▶ **Event:** a subset of the sample space
- ▶ **Probability function  $\Pr$ :** assigns a probability  $\Pr(E)$  to each event  $E$

# PROBABILITY FUNCTION

- ▶ Each outcome in the sample space  $\Omega$  is assigned a probability, which is a number greater or equal to 0.
- ▶ All probabilities of outcomes in  $\Omega$  must add up to 1.



$$\frac{1}{4}$$



$$\frac{1}{4}$$



$$\frac{1}{4}$$



$$\frac{1}{4}$$

- ▶ Probability of event  $E$ , denoted  $\text{Pr}(E)$ , is the sum of probabilities of all outcomes in  $E$ .

## PROBABILITY $\Pr(E)$

- ▶  $\Pr(E)$  = Probability of the event  $E$
- ▶  $\Pr(E)$  is usually given by a "probability fraction"
  - ▶ The **numerator** represents the number of desired outcomes (favorable outcomes)
  - ▶ The **denominator** represents the total number of possible outcomes (size of the sample space).

## PROBABILITY FUNCTION

- ▶ Assigns a probability  $\Pr(E)$  to each event  $E$
- ▶ Experiment: toss a coin

$$\Omega = \{H, T\}$$

# PROBABILITY FUNCTION

- Assigns a probability  $\Pr(E)$  to each event  $E$
- Experiment: toss a coin

$$\Omega = \{H, T\}$$

| $E$      | $\emptyset$ | $\{H\}$ | $\{T\}$ | $\{H, T\}$ |
|----------|-------------|---------|---------|------------|
| $\Pr(E)$ |             |         |         |            |

# PROBABILITY FUNCTION

- Assigns a probability  $\Pr(E)$  to each event  $E$
- Experiment: toss a coin

$$\Omega = \{H, T\}$$

| $E$      | $\emptyset$ | $\{H\}$ | $\{T\}$ | $\{H, T\}$ |
|----------|-------------|---------|---------|------------|
| $\Pr(E)$ |             |         |         |            |

# PROBABILITY FUNCTION

- Assigns a probability  $\Pr(E)$  to each event  $E$
- Experiment: toss a coin

$$\Omega = \{H, T\}$$

Symmetry:

Each outcome is equally likely

| $E$      | $\emptyset$ | $\{H\}$       | $\{T\}$       | $\{H, T\}$ |
|----------|-------------|---------------|---------------|------------|
| $\Pr(E)$ |             | $\frac{1}{2}$ | $\frac{1}{2}$ |            |



# PROBABILITY FUNCTION

- Assigns a probability  $\Pr(E)$  to each event  $E$
- Experiment: toss a coin

$$\Omega = \{H, T\}$$

| $E$      | $\emptyset$ | $\{H\}$       | $\{T\}$       | $\{H, T\}$ |
|----------|-------------|---------------|---------------|------------|
| $\Pr(E)$ |             | $\frac{1}{2}$ | $\frac{1}{2}$ |            |



# PROBABILITY FUNCTION

- Assigns a probability  $\Pr(E)$  to each event  $E$
- Experiment: toss a coin

$$\Omega = \{H, T\}$$

| $E$      | $\emptyset$ | $\{H\}$       | $\{T\}$       | $\{H, T\}$ |
|----------|-------------|---------------|---------------|------------|
| $\Pr(E)$ | 0           | $\frac{1}{2}$ | $\frac{1}{2}$ |            |

# PROBABILITY FUNCTION

- Assigns a probability  $\Pr(E)$  to each event  $E$
- Experiment: toss a coin

$$\Omega = \{H, T\}$$

| $E$      | $\emptyset$ | $\{H\}$       | $\{T\}$       | $\{H, T\}$ |
|----------|-------------|---------------|---------------|------------|
| $\Pr(E)$ | 0           | $\frac{1}{2}$ | $\frac{1}{2}$ |            |

# PROBABILITY FUNCTION

- Assigns a probability  $\Pr(E)$  to each event  $E$
- Experiment: toss a coin

$$\Omega = \{H, T\}$$

| $E$      | $\emptyset$ | $\{H\}$       | $\{T\}$       | $\{H, T\}$ |
|----------|-------------|---------------|---------------|------------|
| $\Pr(E)$ | 0           | $\frac{1}{2}$ | $\frac{1}{2}$ | 1          |

# PROBABILITY FUNCTION

- Assigns a probability  $\Pr(E)$  to each event  $E$
- Experiment: toss a coin

$$\Omega = \{H, T\}$$

| $E$      | $\emptyset$ | $\{H\}$       | $\{T\}$       | $\{H, T\}$ |
|----------|-------------|---------------|---------------|------------|
| $\Pr(E)$ | 0           | $\frac{1}{2}$ | $\frac{1}{2}$ | 1          |

Note: Instead of

$$\Pr(\{H\})$$

we usually write

$$\Pr(H)$$

when we have just one outcome

Compact description:  $\Pr(E) = \frac{|E|}{|\Omega|}$  for all events  $E \subseteq \Omega$

## PROBABILITY FUNCTION

- ▶ Assigns a probability  $\Pr(E)$  to each event  $E$
- ▶ Experiment: roll a die

$$\Omega = \{1,2,3,4,5,6\}$$

# PROBABILITY FUNCTION

- Assigns a probability  $\Pr(E)$  to each event  $E$
- Experiment: roll a die

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

**Symmetry:**

Each outcome is equally likely

$$\Pr(\{1\}) = \Pr(\{2\}) = \dots = \Pr(\{6\}) = \frac{1}{6}$$

# PROBABILITY FUNCTION

- Assigns a probability  $\Pr(E)$  to each event  $E$
- Experiment: roll a die

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

Symmetry:

Each outcome is equally likely

$$\Pr(\{1\}) = \Pr(\{2\}) = \dots = \Pr(\{6\}) = \frac{1}{6}$$

For larger events  $E$ ,  $\Pr(E)$  is the sum of the probabilities of the outcomes in  $S$

$$\Pr(\text{"even"}) = \Pr(\{2, 4, 6\}) = \Pr(\{2\}) + \Pr(\{4\}) + \Pr(\{6\}) = \frac{3}{6} = \frac{1}{2}$$

*Handwritten red text above the equation:  $\Pr(\{2\} \cup \{4\} \cup \{6\})$*



# PROBABILITY FUNCTION

- Assigns a probability  $\Pr(E)$  to each event  $E$
- Experiment: roll a die

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

Symmetry:

Each outcome is equally likely

$$\Pr(\{1\}) = \Pr(\{2\}) = \dots = \Pr(\{6\}) = \frac{1}{6}$$

For larger events  $E$ ,  $\Pr(E)$  is the sum of the probabilities of the outcomes in  $E$

$$\Pr(E) = \frac{|E|}{|\Omega|} \text{ for all events } E \subseteq \Omega$$



## PROBABILITY FUNCTION

Symmetry:

Each outcome is equally likely

- ▶ Assigns a probability  $\Pr(E)$  to each event  $E$
- ▶ Experiment: toss a coin 3 times

$$\Omega = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$$

$$\Pr(\{HHH\}) = \Pr(\{HHT\}) = \dots = \Pr(\{TTT\}) = \frac{1}{8}$$

# PROBABILITY FUNCTION

Symmetry:

Each outcome is equally likely

- ▶ Assigns a probability  $\Pr(E)$  to each event  $E$
- ▶ Experiment: toss a coin 3 times

$$\Omega = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$$

$$\Pr(\{HHH\}) = \Pr(\{HHT\}) = \dots = \Pr(\{TTT\}) = \frac{1}{8}$$

For larger events  ~~$E$~~ ,  $\Pr(\textcolor{red}{E})$  is the sum of the probabilities of the outcomes in  ~~$E$~~

$$\Pr(E) = \frac{|E|}{|\Omega|} \text{ for all events } E \subseteq \Omega$$

## PROBABILITY FUNCTION

- ▶ Assigns a probability  $\Pr(E)$  to each event  $E$
- ▶ Experiment: toss a coin 3 times

$$\Omega = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$$

- ▶ The coin tosses are **independent** (the outcome of a toss does not change the probability that a different toss is H or T)

$$\Pr(\{HHT\}) = \Pr("1^{st} \text{ toss is } H" \cap "2^{nd} \text{ toss is } H" \cap "3^{rd} \text{ toss is } T")$$

## PROBABILITY FUNCTION

- ▶ Assigns a probability  $\Pr(E)$  to each event  $E$
- ▶ Experiment: toss a coin 3 times

$$\Omega = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$$

- ▶ The coin tosses are **independent** (the outcome of a toss does not change the probability that a different toss is H or T)

$$\begin{aligned} \Pr(\{HHT\}) &= \Pr("1^{st} \text{ toss is } H" \cap "2^{nd} \text{ toss is } H" \cap "3^{rd} \text{ toss is } T") \\ &= \Pr("1^{st} \text{ toss is } H") \cdot \Pr("2^{nd} \text{ toss is } H") \cdot \Pr("3^{rd} \text{ toss is } T") \end{aligned}$$

product rule for independent events (will justify it later)

## PROBABILITY FUNCTION

- ▶ Assigns a probability  $\Pr(E)$  to each event  $E$
- ▶ Experiment: toss a coin 3 times  
 $\Omega = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$
- ▶ The coin tosses are **independent** (the outcome of a toss does not change the probability that a different toss is H or T)

$$\begin{aligned}\Pr(\{HHT\}) &= \Pr("1^{\text{st}} \text{ toss is H} \cap "2^{\text{nd}} \text{ toss is H} \cap "3^{\text{rd}} \text{ toss is T}") \\ &= \Pr("1^{\text{st}} \text{ toss is H}") \cdot \Pr("2^{\text{nd}} \text{ toss is H}") \cdot \Pr("3^{\text{rd}} \text{ toss is T}") \\ &= \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}\end{aligned}$$

# PROBABILITY FUNCTION

Not symmetric:  
Outcomes are **not** equally likely

- Assigns a probability  $\Pr(S)$  to each event  $E$
- Experiment: toss a coin until heads for the first time

$$\Omega = \underbrace{\{TT \dots T H : i \in \mathbb{N}\}}_{i \text{ times}} \quad \Pr(H) = \frac{1}{2}$$

$$\Pr(TH) = \Pr(H) \cdot \Pr(T) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

The coin tosses are **independent** (the outcome of a toss does not change the probability that a different toss is H or T)

$$\Pr(\underbrace{\{TTT \dots T H\}}_{i \text{ tails}}) = \left(\frac{1}{2}\right)^i \cdot \frac{1}{2} = \left(\frac{1}{2}\right)^{i+1}, \quad i \in \mathbb{N}$$

# PROBABILITY FUNCTION

What principles did we use to come up with those probability functions?

- ▶ **Symmetry:** each outcome of the coin toss (or die roll) is equally likely
- ▶ The probability of each outcome is a number between 0 and 1
- ▶ **Additivity:** for events with more than one outcome, the probability of the event is the sum of the probabilities of its outcomes

**Note:** some experiments are not symmetric (toss a coin until H)



# PROBABILITY FUNCTION: AXIOMS

- ▶ Probability function assigns a probability to each event
- ▶ For a function to be a probability function, it must satisfy the following properties, called **axioms of probability**
- ▶ **Non-negativity:**  $Pr(E) \geq 0$  for all events  $E \subseteq \Omega$
- ▶ **Additivity:** if  $A$  and  $B$  are **disjoint** events then

$$Pr(A \cup B) = Pr(A) + Pr(B)$$

- ▶ **Normalization:**  $Pr(\Omega) = 1$



# PROBABILITY FUNCTION: AXIOMS

- ▶ **Non-negativity:**  $Pr(E) \geq 0$  for all events  $E \subseteq \Omega$
- ▶ **Additivity:** if A and B are **disjoint** events then

$$Pr(A \cup B) = Pr(A) + Pr(B)$$

- ▶ If  $A_1, A_2, \dots$  are **disjoint** events then

$$Pr(A_1 \cup A_2 \cup \dots) = Pr(A_1) + Pr(A_2) + \dots$$

- ▶ **Normalization:**  $Pr(\Omega) = 1$

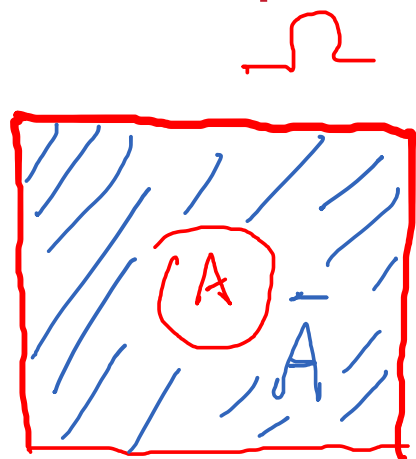
Do we need to add  $0 \leq Pr(E) \leq 1$ ?

# PROBABILITY RULES

(Proof on the slide)

- ▶ We can derive many useful properties from the axioms

- ▶ Complement rule:  $Pr(\bar{A}) = 1 - Pr(A)$



$$\Omega = A \cup \bar{A}$$

$$Pr(\Omega) = Pr(A \cup \bar{A})$$

$$1 = Pr(A) + Pr(\bar{A})$$

$$Pr(\bar{A}) = 1 - Pr(A)$$

□

definition of  $\Omega$

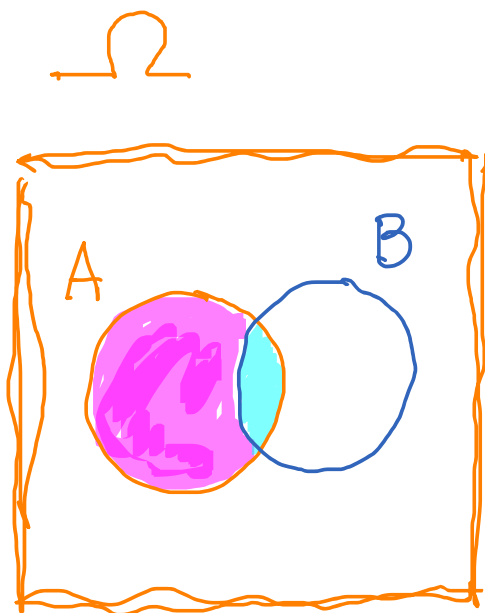
additivity  
axiom

normalization  
axiom

# PROBABILITY RULES

(Proof on the slide)

- ▶ We can derive many useful properties from the axioms
- ▶ Difference rule:  $Pr(A \setminus B) = Pr(A) - Pr(A \cap B)$



$$A = (A \setminus B) \cup (A \cap B)$$

$$Pr(A) = Pr(A \setminus B) + Pr(A \cap B)$$

additivity  
axiom

$$Pr(A \setminus B) = Pr(A) - Pr(A \cap B)$$

□

# PROBABILITY RULES

(Proof on the slide)

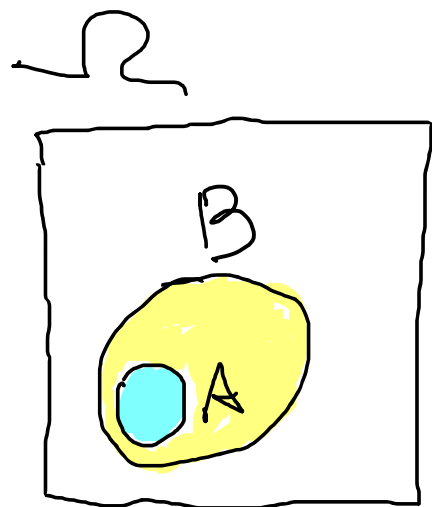
- ▶ We can derive many useful properties from the axioms
- ▶ Monotonicity rule:  $A \subseteq B \Rightarrow \Pr(A) \leq \Pr(B)$

$$\Pr(B) = \Pr(A \cup (B \setminus A))$$

$$\Pr(B) = \Pr(A) + \underline{\Pr(B \setminus A)}$$

Note that  $\Pr(B \setminus A) \geq 0$

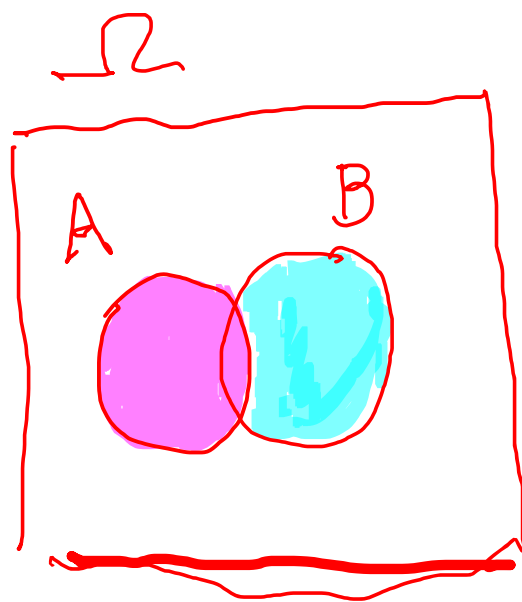
$$\Pr(B) \geq \Pr(A)$$



# PROBABILITY RULES

(Proof on the slide)

- ▶ We can derive many useful properties from the axioms
- ▶ Inclusion-exclusion rule:  $Pr(A \cup B) = Pr(A) + Pr(B) - Pr(A \cap B)$



$$Pr(A \cup B) = Pr(A \cup (B \setminus A))$$

$$Pr(A \cup B) = Pr(A) + \underbrace{Pr(B \setminus A)}$$

$$Pr(A \cup B) = Pr(A) + Pr(B) - Pr(A \cap B)$$

□

# PROBABILITY RULES

(Proof on the slide)

- ▶ We can derive many useful properties from the axioms
- ▶ Union-bound inequality:  $Pr(A \cup B) \leq Pr(A) + Pr(B)$

$$Pr(A \cup B) = Pr(A) + Pr(B) - Pr(A \cap B)$$

Note that  $Pr(A \cap B) \geq 0$ , so

$$Pr(A \cup B) \leq Pr(A) + Pr(B) \quad \square$$

## ASK THE AUDIENCE

- 50 students showed up to class today
- 20 are Red Sox fans (R), 25 are Patriots fans (P)
- Prof. Tiago chooses a student uniformly at random
- What is the range of possible values for  $p = \Pr(R \cup P)$ ?

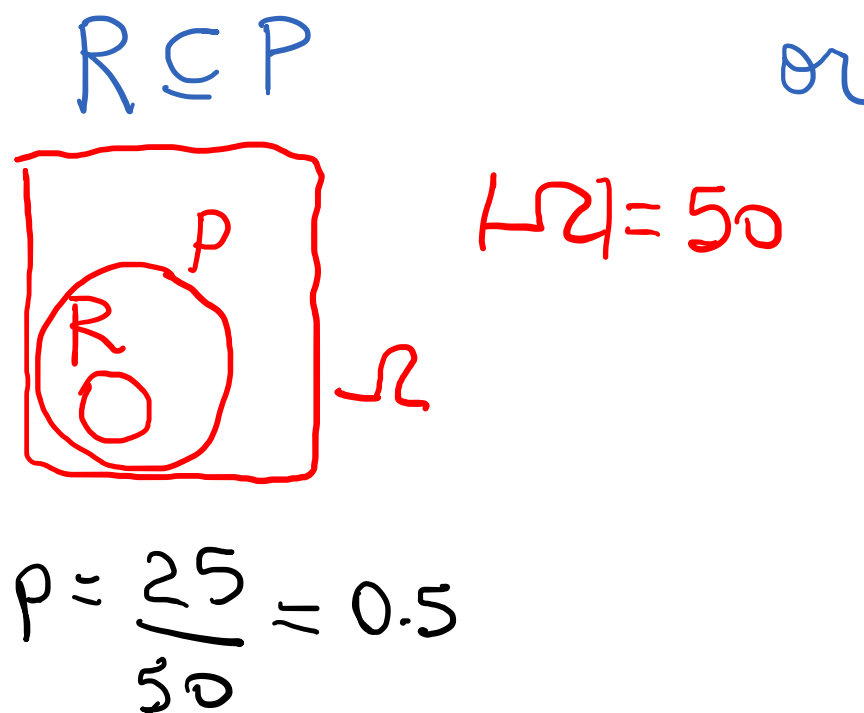
A.  $p \geq 0.5$

B.  $p \leq 0.4$

C.  $0.4 \leq p \leq 0.9$

~~D.  $0.5 \leq p \leq 0.9$~~

~~E.  $0.4 \leq p \leq 0.5$~~



$$p = \frac{45}{50} = \frac{9}{10} = 0.9$$